

Polypharmacy Drug-Drug Interaction (DDI) Prediction System

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1. Abstract

This study presents a machine learning system that predicts the risk of adverse interactions that may arise from the simultaneous use of multiple drugs. Using the TWOSIDES dataset, a binary classification model based on the molecular structure of drug pairs (Morgan Fingerprint) was developed; Logistic Regression, Random Forest, and XGBoost were compared, and Logistic Regression, which yielded the highest Recall and ROC-AUC values, was selected as the final model. The system has been extended, through pairwise combination decomposition, to evaluate polypharmacy scenarios involving up to 10 drugs, and has been presented as an interactive application via Streamlit.

2. Introduction

Polypharmacy is a clinically preventable risk factor that threatens patient safety, particularly in elderly and chronically ill patient populations. Early prediction can provide input to pre-prescription decision support systems. The input, output, and task type information for this project is as follows:

- Input: Chemical structure (SMILES representation) and identity information of the drugs.
- Output: Whether an interaction event occurs when the drugs (up to 10) are used together
- Task type: Supervised classification.

3. General System Architecture

The developed system supports entering up to 10 drugs in the user interface built with Streamlit. The model analyzes all entered drugs by breaking them down into pairwise combinations. The system runs a separate interaction prediction for each of these pairs and presents a comprehensive combination report.

Consequently, the model cannot calculate the synergistic effects created by higher-order combinations (taking 3, 4, or more drugs at the same time). For example, all pairwise combinations (A-B, A-C, B-C) of 3 drugs entered into the system may come out as safe; however, when these 3 drugs are taken into the body simultaneously, a completely different and unforeseeable adverse effect may be triggered. Due to the structure of the dataset, the model does not have the ability to predict this kind of collective interaction.

Below, in order to test the clinical accuracy of the developed Streamlit interface and the Logistic Regression-based prediction model, a 3-drug scenario consisting of Warfarin, Aspirin, and Ibuprofen a combination considered extremely risky to use together in the medical literature has been simulated.

The screenshot shows a web interface titled "Medication List Management". At the top, there is a slider control labeled "Select number of medications to analyze (2-10):" with the value set to 3. Below this, there are three sections for medication entry:

- Medication #1:** "Enter Drug Name #1" with the input "Warfarin". Below the input is a green checkmark and the text "✓ 'Warfarin' identified successfully."
- Medication #2:** "Enter Drug Name #2" with the input "Ibuprofen". Below the input is a green checkmark and the text "✓ 'Ibuprofen' identified successfully."
- Medication #3:** "Enter Drug Name #3" with the input "Aspirin". Below the input is a green checkmark and the text "✓ 'Aspirin' identified successfully."

Since the system is structurally optimized to predict (*pairwise*) interactions, this 3-drug combination has been isolated in the background and subjected to analysis as 3 separate pairwise pairs.

As can be seen in the screenshot below, the model has successfully interpreted the chemical and clinical features it learned from the TWOSIDES dataset (via SMILES inputs), precisely revealing the severity of the clinical risk.

As clinical assistant support, how the risk decision was reached is shown via the SHAP model, through the Explainable AI Analysis feature.

The screenshot shows a web interface titled "Analyze Interaction Risks". It displays three pairwise interaction risk analyses, each in a light blue box with a red header:

- Interaction Risk Detected: Warfarin + Aspirin**
Risk Status: HIGH RISK | Interaction Probability: 99.99%
Below the box is a button labeled "> Explainable AI (SHAP) Analysis for Warfarin + Aspirin".
- Interaction Risk Detected: Warfarin + Ibuprofen**
Risk Status: HIGH RISK | Interaction Probability: 99.98%
Below the box is a button labeled "> Explainable AI (SHAP) Analysis for Warfarin + Ibuprofen".
- Interaction Risk Detected: Aspirin + Ibuprofen**
Risk Status: HIGH RISK | Interaction Probability: 99.94%
Below the box is a button labeled "> Explainable AI (SHAP) Analysis for Aspirin + Ibuprofen".

4. Data and Methods Used

4.1 Data Selection

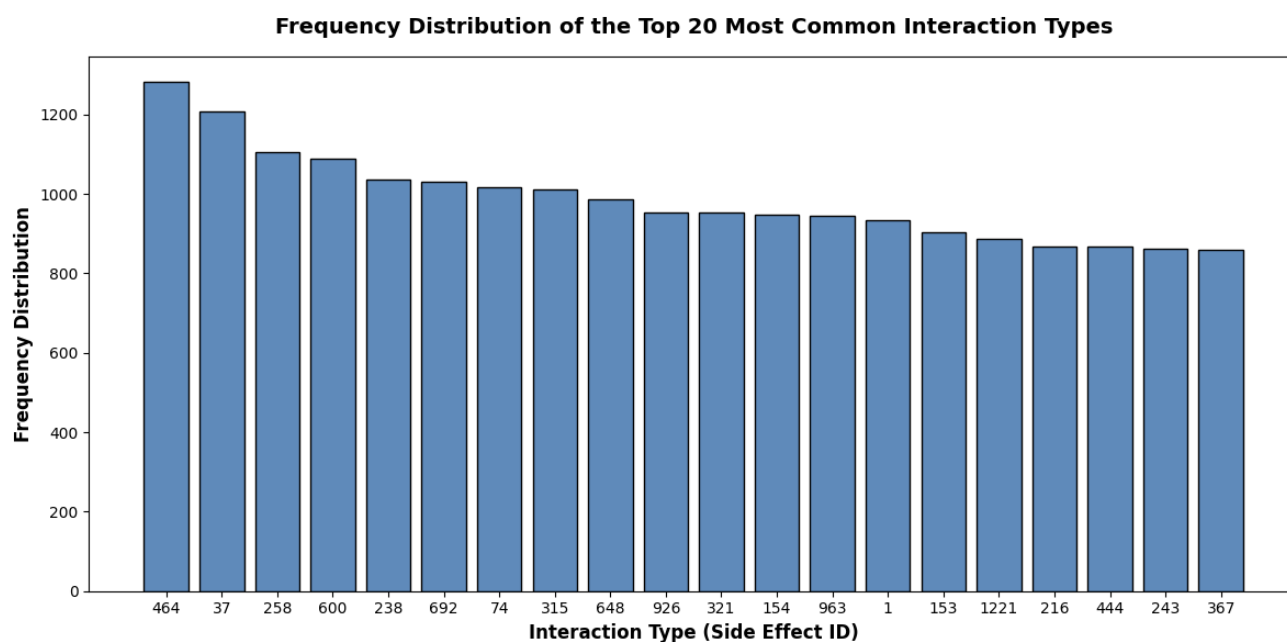
The dataset commonly used in DDI studies in the literature is the DrugBank library. However, DrugBank's academic download program has been suspended since May 2026 and is not expected to reopen in the short term; therefore, the project was carried out using the TWOSIDES dataset.

The TWOSIDES dataset was accessed via the PyTDC library and contains a total of 4,649,441 drug pair records. Due to computational constraints, a random subsample (N=200,000) was selected while preserving class balance, and all modeling was carried out on this subsample.

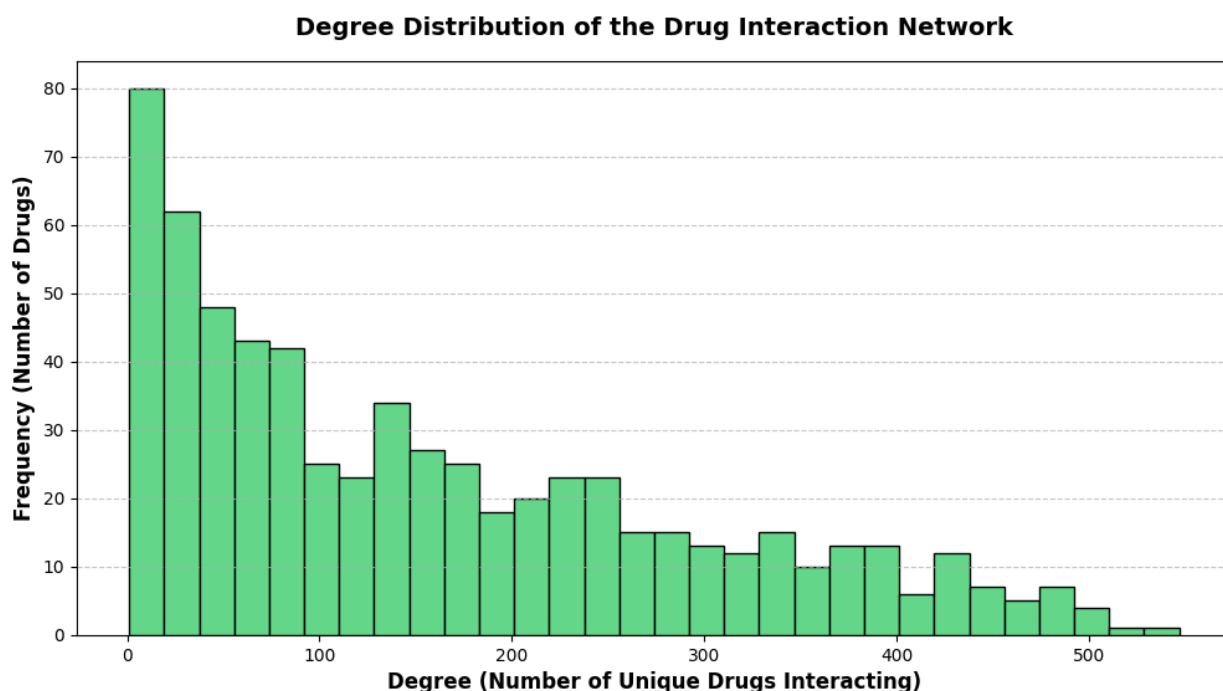
4.2 Data Processing

First, data quality was reviewed and missing or incorrectly entered records were checked. As a result of the EDA, no empty data or undefined drugs were observed.

When the frequency distribution of the Top 20 Most Common Interaction Types below is examined, it is observed that there is no sharp divide between classes and that the frequencies show a smooth transition. This indicates that, with respect to interaction types, the model will not experience serious class imbalance during training, and that the model's development of bias while learning these classes will be minimized.



On the other hand, when the Degree Distribution graph of the Drug Interaction Network below is examined, it is seen that the dataset has a scale-free network structure. While a small number of likely popular drugs have a very high number (500+) of interactions, the vast majority of drugs have fewer interactions. This creates an imbalance on the input side. The model will learn the chemical structures of frequently interacting (high-degree) drugs much more easily and successfully, because there is a large amount of data on them. However, since it will have few examples for drugs with very few interactions, the model may struggle to predict the interactions of these rare drugs.



Next, a Morgan Fingerprint was extracted for each drug using the RDKit library.

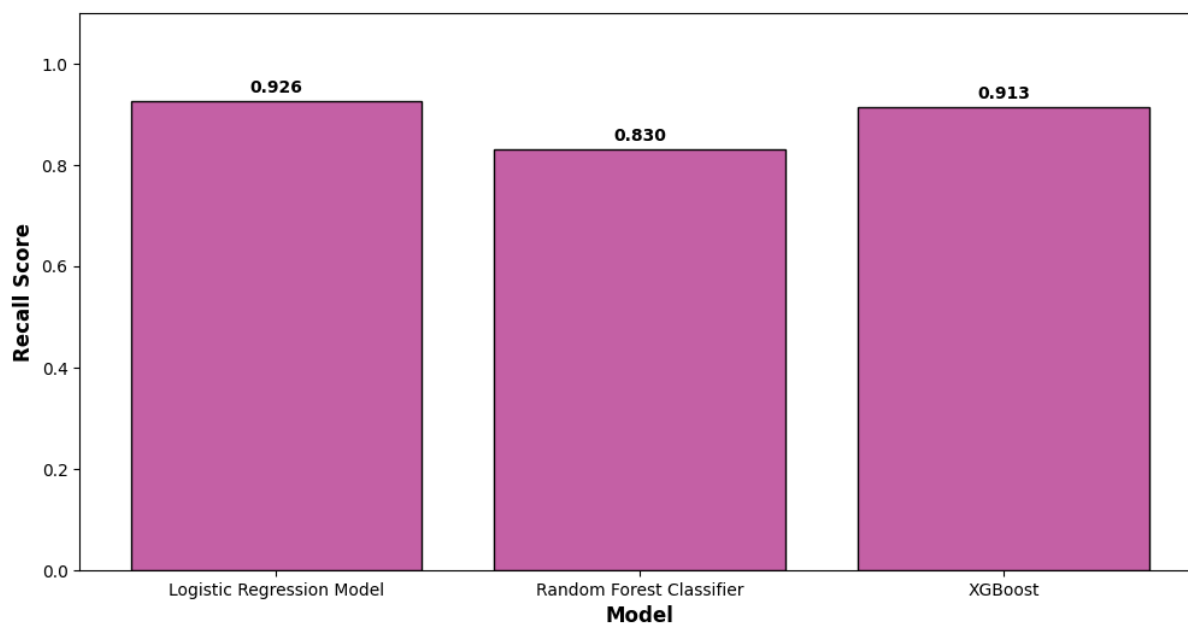
The TWOSIDES dataset only contains drug pairs that actually interact, but for a model to learn the interaction-present/absent distinction, both positive and negative samples are required. Without negative samples, the model always learns to say "interaction present" because it has never seen an absent example.

Since there are no negative samples in the dataset, negative samples are generated based on the standard approach in the literature. If two drugs never appear together in the dataset, there is presumably no known interaction between them. This assumption is not 100% accurate; it is merely a practical approach and is, in fact, a limitation. Negative samples were synthetically generated so that the positive:negative ratio is 1:1.

Finally, the fingerprints of the two drugs were combined into a single 4096-dimensional vector using the concatenation (end-to-end joining) method. Concatenation was preferred because it is the most commonly used method in DDI prediction in the literature and minimizes information loss.

5. Modeling

In order to obtain the most accurate result, the dataset was tested on three different classification models (Logistic Regression, Random Forest Classifier, XGBoost), and the model yielding the highest Recall score was accepted as the baseline. All three models were compared using the same train/test split (0.8 / 0.2) and fingerprint data combined via concatenation. Since the clinical cost of a missed interaction (False Negative) is high, Recall was the primary criterion in model selection, although Precision and ROC-AUC values were also taken into consideration.

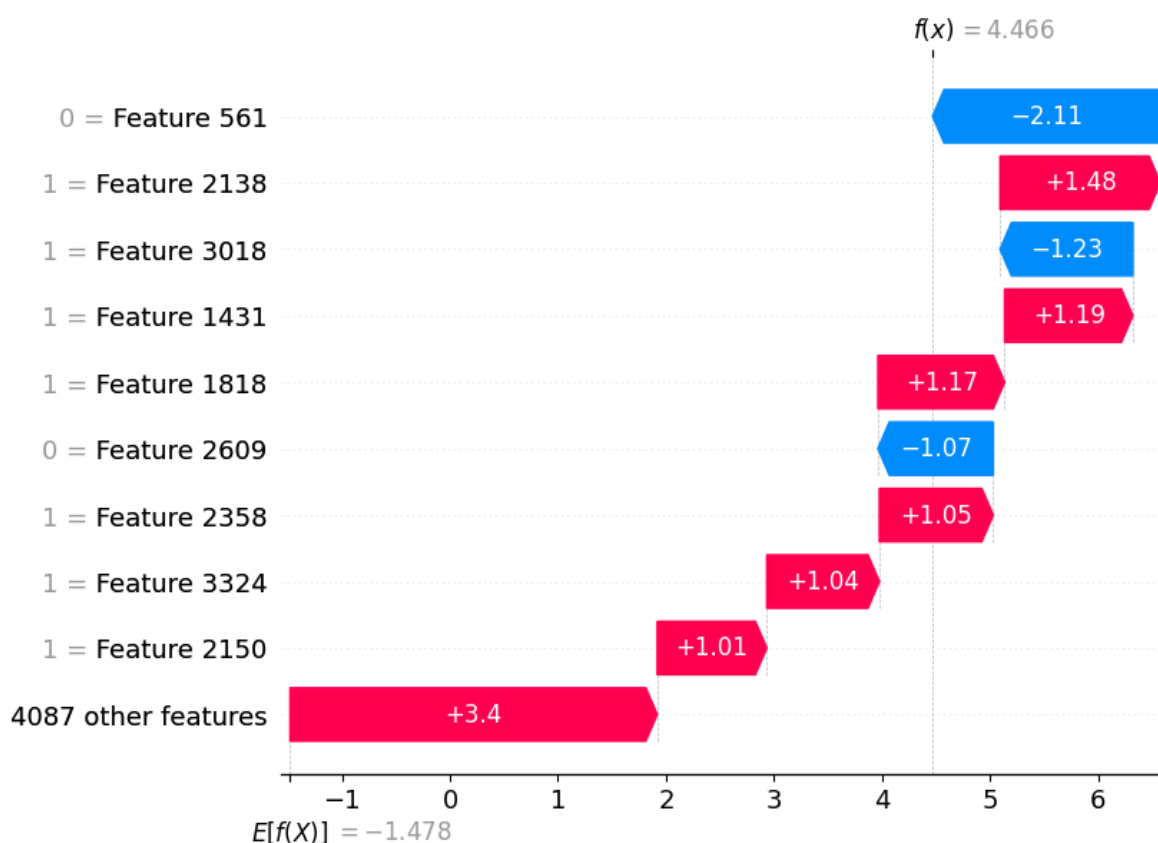


Logistic Regression yielded the highest Recall (0.93) and ROC-AUC (0.98) values, followed by XGBoost (Recall: 0.91) and Random Forest (Recall: 0.83). This result suggests that the high-dimensional and sparse binary structure of the Morgan Fingerprint representation may be more amenable to a linear relationship than to the complex relationships sought by tree-based models. For this reason, Logistic Regression was selected as the final model.

A false positive analysis was also performed on the selected model. The results show that the model correctly classified 91.45% of the negative (no-interaction) test samples, while 8.55% were misclassified as false positives. This error rate is expected in clinical decision-support systems and is consistent with the model's design priority: recall was favored over precision, since missing a real interaction (false negative) is considered more costly than raising a false alarm (false positive).

6. SHAP Interpretability

In order for the model to be considered reliable in a clinical context, it is important that its predictions be explainable. Using the SHAP (SHapley Additive exPlanations) method, it was analyzed which fingerprint bits influenced the model's individual predictions, and in which direction. The fact that fingerprint bits do not directly correspond to chemical substructure names is a limitation of this interpretation.



The waterfall plot above shows, step by step, the model's decision for a randomly selected test example. $E[f(X)] = -0.998$ represents the model's average prediction (reference point) across the entire dataset — that is, the score the model would give if it made a prediction with no specific information. For this particular example, the model produced a score of $f(x) = 4.385$. This is significantly above the average, meaning the model strongly supports the likelihood that this pair interacts. Despite a few reducing (blue) features, the fact that numerous small positive (red) contributions dominate in total has led the model to classify this pair as high risk.

In this drug pair, the absence of Feature 561 (0 = Feature 561) alone produced the strongest negative effect (-2.34) — meaning the absence of this bit somewhat reduces the risk. Nevertheless, the presence of 8 different bits, and especially the combined effect of the 4087 other features (+3.01), has more than offset this single negative effect. As a result, the model produced a high risk score of 4.385, well above the average (-0.998) — meaning it strongly supports the likelihood that these two drugs interact.

7. Limitations

- The suspension of DrugBank's academic access
- The use of a 200,000-row subsample instead of the full dataset, due to RAM constraints
- The reliance of the synthetic negative sampling method on the assumption that "a pair not present in the dataset = no interaction"
- The inability to model emergent (three-or-more-drug) interactions
- The limited direct chemical interpretability of fingerprint bits

8. Conclusion

This study has presented an explainable and extensible prediction system for a clinically significant patient safety problem (polypharmacy), using the TWOSIDES dataset and classical machine learning methods.

The selection of Logistic Regression as the final model, which yielded the highest Recall and ROC-AUC values among the three, stands out as a notable finding, suggesting that linear models can outperform tree-based models on high-dimensional, sparse fingerprint data.